CS215 Open Project Proposal
Team 20: Does Rain Make Onions Cry?
Studying the Effect of Heavy Monsoon Rainfall on Onion Prices in Indian Markets

1. Main Hypothesis / Research Question

Do unusually heavy monsoon rains increase the probability of sharp rises in onion prices in Indian agricultural markets? Excess rainfall may damage crops, disrupt transport, or delay arrivals, leading to sudden increases in market prices. However, onion prices also depend on storage, harvest cycles, and other supply conditions.

Our main null hypothesis (H_0) is that the probability of an onion-price spike is the same after unusually wet and normal-rainfall periods. The alternative (H_1) is that price spikes are more likely after unusually wet periods.

We model the price-spike indicator as a Bernoulli random variable *conditional on rainfall conditions*:

$$S \mid W = w \sim \text{Bernoulli}(p_w), \qquad p_w = P(S = 1 \mid W = w).$$

Our hypotheses are

$$H_0 : p_1 = p_0, \qquad H_1 : p_1 > p_0.$$

Here, $W = 1$ denotes an unusually wet period and $S = 1$ denotes a sharp onion-price increase. Using monthly representative onion prices P_t, we define

$$\Delta_t = 100\frac{P_t - P_{t-1}}{P_{t-1}}, \qquad S_t = \mathbf{1}\{\Delta_t \geq 20\%\}.$$

The 20% spike threshold is preliminary and will be checked using other thresholds. For rainfall, we will tentatively define $W = 1$ as at least 20% above the district-month normal and $W = 0$ as close to normal; unusually dry periods will be excluded from this comparison.

2. Source Papers, Reports and Other Reading

- AGMARKNET data portal and AGMARKNET training presentation: wholesale price data and reporting context.
- India Meteorological Department (IMD): rainfall observations, normals, and monsoon reports.
- Sekhar (2012), *Food Policy*: background on Indian agricultural markets.
- CEDA Agri-Market Data Portal: complementary agricultural market data.
- Further reading on onion-price volatility and monsoon-related supply disruptions will be selected after the feasibility check.

3. Datasets

- **Onion prices:** Daily mandi-level prices from AGMARKNET, with data.gov.in as an access option, aggregated to monthly prices.
- **Rainfall:** District-level monthly rainfall and historical normals from IMD, matched to market locations.
- **Validation:** Check coverage using CEDA and historical AGMARKNET extraction notes.
- **Scope:** Major onion-producing markets, initially in Maharashtra and nearby states, over the longest reliable period available (ideally 2018–2025).

4. Programming Languages / Tools

Python (pandas, NumPy, SciPy, statsmodels, and matplotlib) and R will be used for data cleaning and analysis. Jupyter notebooks and GitHub will make the work reproducible.

5. Analysis Plan and Deliverables

1. **Data cleaning:** Check missing prices, duplicates, inconsistent units, extreme outliers, and rainfall coverage. Match mandis to districts and align the dates.
2. **Conditional probability:** Estimate $p_1 = P(S = 1 \mid W = 1)$ and $p_0 = P(S = 1 \mid W = 0)$, and compare the estimates and their confidence intervals.
3. **Conditional Bernoulli model (primary analysis):** Treat $S \mid W = w$ as Bernoulli(p_w) and test $H_0 : p_1 = p_0$ against $H_1 : p_1 > p_0$ to assess whether price spikes are more likely following unusually wet periods.
4. **Two-sample t-test (secondary analysis):** Compare mean percentage price changes in unusually wet and normal-rainfall periods using a standard pooled two-sample t-test, provided its independence and equal-variance assumptions are reasonable.
5. **Distribution analysis:** Compare the spread of price changes, volatility, and unusually large increases using histograms and box plots.
6. **Delayed effects and robustness:** Check rainfall in the same month and one or two previous months. Repeat the analysis with different spike thresholds and account for market and seasonal differences.

Main challenge: Rainfall may influence onion prices with a delay. Storage, harvest cycles, transportation, and other supply disruptions may also affect prices. Repeated observations from a market are not fully independent, so uncertainty estimates should account for clustering. This study can identify associations, but not establish that rainfall alone causes price increases.

6. Logistics and Work Distribution

Team members will divide data collection, cleaning, exploratory analysis, statistical testing, and report writing. Code and progress updates will be maintained in the Team 20 GitHub repository. The team schedule spreadsheet will track tasks and deadlines.

7. LLM Use / Conversation Links

Any AI assistance used for drafting or coding will be documented according to course requirements.

8. Help Requested from the Teaching Team

We would appreciate guidance on defining unusually heavy rainfall, choosing a meaningful price-spike threshold, accounting for delayed effects, and obtaining reliable historical data from AGMARKNET and IMD.